Squadron Koller: Team and Roles

Squadron Koller: Team and Roles (Mission 02)

Mission: 02 FST Dialogue Systems, Lovelace's Objection & Noisy Channels
Course: Conversational AI - Winter Semester 2026/27
Repository: https://github.com/Unisvet/couse-c-ai/tree/main/mission-02

1. Team Roster and Station Assignments

The following table outlines the operational roles for Mission 02.

Name	Squadron Station	Mission 02 Role
[Antoun Al Salameh 5019770]	Data & Payload Specialist	Part 1: Historical Archivist & Probing
[Antoun Al Salameh 5019770]	Communications & Protocol Officer	Part 2: State Machine & Constraints
[Ragni Chawala 5012421]	Security & Ethics Sentinel	Part 3: Hazard & Entropy Auditor
[Ragni Chawala 5012421]	Systems Architect	Part 4: Core Simulation Builder
[Ragni Chawala 5012421]	AI Security Officer	Mission 02 EU AI Act Compliance

2. Detailed Role Responsibilities & Work Distribution

Data & Payload Specialist

Responsible for the theoretical groundwork surrounding the philosophy of mind and artificial intelligence. This officer must deeply analyze Ada Lovelace's Note G regarding the Analytical Engine's inability to originate thought, contrasting it with Alan Turing's Argument from Consciousness from his 1950 paper. The officer will design an Anti-Turing Diagnostic Protocol consisting of 5 highly specific questions designed to trap a modern LLM into revealing its stochastic nature. They will conduct the live LLM evaluation and document the verbatim responses.

Communications & Protocol Officer

Tasked with the deterministic architecture of the dialogue system. This officer must define the formal Mealy Finite State Transducer (FST) that powers AURA Subsystem 02. They are responsible for mapping the 5 required states (INIT, GREETING, COLLECT_SLOTS, CONFIRMATION, TERMINAL/FALLBACK) and defining the strict regex extraction rules for destination, speed, and clearance code. Furthermore, they will formulate the Shannon Noisy Channel constraint strategy, ensuring that responses

never exceed 7 words and always inject a mandatory noise token from the designated vocabulary.

Security & Ethics Sentinel

Serves as the red-team auditor for the FST system. This officer must design and execute 5 rigorous stress tests against the deterministic FST, employing adversarial tactics such as over-provisioning, backtracking, and off-topic hallucination induction. They will mathematically and qualitatively evaluate the degradation of mutual information (Semantic Drift) caused by the Noisy Channel constraints, assessing the hallucination risks when deterministic symbolic outputs are forcibly injected with stochastic noise.

Systems Architect

The lead developer responsible for synthesizing the theoretical models into a functional Python simulator. This officer will build the `FSTDialogueManager` class, integrate the regex slot memory, enforce the Noisy Channel token limitations, and execute the 10-turn Noisy Interrogation Transcript. They must ensure the code runs flawlessly in the Antigravity IDE and handles the git repository layout.

AI Security Officer (Compliance)

Responsible for evaluating the new AURA Subsystem 02 against the EU AI Act (Regulation 2024/1689). They must determine if the forced noise injection constitutes a prohibited manipulative practice and ensure transparency obligations (Article 50) are met even within a deterministic framework.